

STATEMENT OF WORK

Title: Repair and Calibration of Joint Combined Aircrew System Tester (JCAST)



National Aeronautics and Space Administration
Armstrong Flight Research Center
Edwards, California

Table of Contents

- 1.0 [Background](#)
 - 2.0 [Objectives](#)
 - 3.0 [Requirements](#)
 - 4.0 [Deliverables](#)
 - 5.0 [Period of Performance](#)
 - 6.0 [Travel](#)
 - 7.0 [Other Special Requirements and Considerations](#)
-

1.0 Background

AFRC utilizes the Joint Combined Aircrew System Tester (JCAST) to support life support testing requirements for programs such as the Low Boom Flight Demonstrator. The JCAST is used in conjunction with its associated subcomponents (ACAST, Valve Test Accessory, Relief Valve Test Accessory), and is essential for the proper testing of CRU-122 oxygen regulators, pressure suits, g-garments, and integrated aircrew life support systems.

2.0 Objectives

The objective of this requirement is to ensure the full restoration, calibration, and verification of the AFRC JCAST system to OEM standards so that accurate, safe, and repeatable life support testing can be performed in support of flight operations

3.0 Requirements

The contractor shall:

- Conduct full diagnostic evaluation of the JCAST unit and all accessories.
- Perform all necessary repairs using OEM parts and proprietary procedures.
- Calibrate the JCAST and subcomponents using the OEM technical publication
- Perform functional testing to validate leak rates, pressure profiles, and performance criteria as required for aircrew ensemble testing.
- Provide a certification of calibration traceable to Chemring-approved standards.
- Ensure all work complies with applicable Air Force Technical Orders (T.O. 15X6-4-8-2, T.O. 33D2-10-141-1). 2
- Return equipment in ready-for-use condition, packaged for safe transport.
- Complete repairs within Chemring's standard 14–16 week timeline from receipt of equipment.

4.0 Deliverables

Contractor shall provide:

- Inspection results and diagnostic report
- Calibration certificate(s)
- Repair summary documentation
- Verification and functional test results
- Proper packaging and shipment back to AFRC
- All required documentation demonstrating equipment acceptance and completion of work

5.0 Period of Performance

The period of performance is 12 months from award date. This allows for repair cycles, calibration lead time, shipping, and administrative requirements.

6.0 Travel

Not applicable

7.0 Other Special Requirements and Considerations

To be determined individually for each piece of equipment.

REVISIONS

Revision Number	Date	Brief Summary of Changes
Base		
1		
2		
3		
4		
5		
6		
7		
8		